X(3872) Analysis Update

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Training selection: PbPb 2023

Common selection: MC and DATA

Variable	Selection before ML
`Bpt`	$15 < \texttt{Bpt} < 50$ GeV
`By`	$\lvert\texttt{By}\rvert < 1.6$
`BQvalue`	$\texttt{BQvalue} < 0.15$ GeV
`Btrk2dR`	$\texttt{Btrk2dR} < 0.35$

Signal: X MC
All pthat components; `pThatreweight`.
No additional mass window in training.

Background: DATA sidebands
$3.80 < m < 3.85$ or $3.89 < m < 3.94$ GeV.
Unit event weights.

Variable selection from two to nine inputs

Allowed pool: 19 inputs

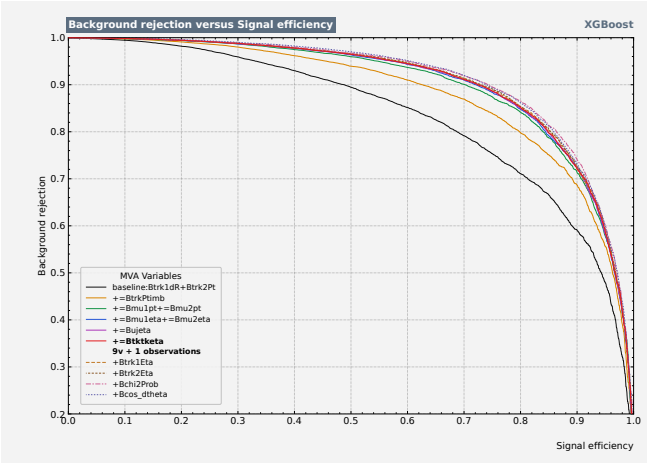
`Btrk1dR`, `Btrk2dR`
`Btrk1Pt`, `Btrk2Pt`, `BtrkPtimb`
`Bmu1pt`, `Bmu2pt`
`Bmu1eta`, `Bmu2eta`
`Btrk1Eta`, `Btrk2Eta`
`Bujeta`, `Btktkpt`, `Btktketa`
`Bchi2Prob`, `BujvProb`, `BtktkvProb`
`Balpha`, `Bcos_dtheta`

Start point: two variables

`Btrk1dR` + `Btrk2Pt`

Add according to AUC
Evaluate additions from the allowed pool.
Rank by pooled train + test AUC.
Select the next variable combination.

AUC: the new two- to nine-variable route



Start point

Btrk1dR

Btrk2Pt

Endpoint: selected variables

Btrk1dR, Btrk2Pt

BtrkPtimb

Bmu1pt, Bmu2pt

Bmu1eta, Bmu2eta

Bujeta, Btktketa

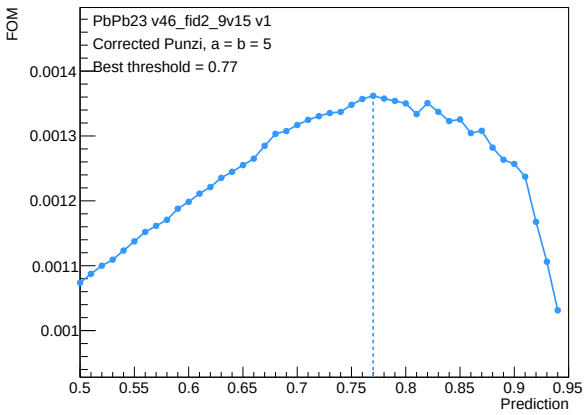
Punzi FOM: empirical corrected form

Signal: $(3.84, 3.90)$ GeV; sidebands: $(3.75, 3.84) \cup (3.90, 3.99)$ GeV.

$$\mathrm{FOM}_{\mathrm{corr}}(c) = \frac{\epsilon_X(c)}{a^2/8 + 9b^2/13 + a\sqrt{B(c)} + \frac{b}{2}\sqrt{b^2 + 4a\sqrt{B(c)} + 4B(c)}}, \qquad a = b = 5.$$

- $B(c)$: DATA sideband count after the score cut, divided by 3.
- $\epsilon_X(c)$: weighted X MC efficiency for score $> c$, within $3.62 < m < 4.00$ GeV.
- MC weight: `pThatreweight`; MC window factor fixed to 1.
- a: discovery threshold; b: target power parameter.

Nine-variable model: Punzi FOM versus score

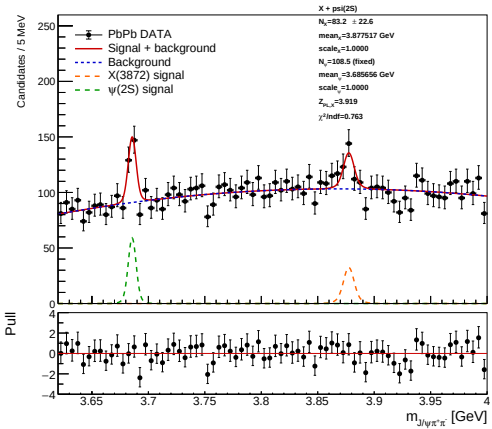


Empirical corrected Punzi
$a = b = 5$.

Score: 0.50–0.95
Step: 0.01

Grid maximum
score > 0.77
FOM $= 1.361820 \times 10^{-3}$

Nine-variable model: two-peak fit at the Punzi optimum



score > 0.77

$N_X = 83.20 \pm 22.64$
$N_{\psi(2S)} = 108.53$ (fixed)

$\sqrt{q_0} = 3.919$

MC double-Gaussian widths fixed.
2nd-order Chebyshev background.

Thanks for listening!

Reporter: Lekai Yao

2026.9.24

Five variables, all pthat: training

Common training selection

Variable	Selection before ML
Bpt	$15 < \texttt{Bpt} < 50$ GeV
By	$\lvert\texttt{By}\rvert < 1.6$
BQvalue	$\texttt{BQvalue} < 0.15$ GeV
Btrk2dR	$\texttt{Btrk2dR} < 0.35$

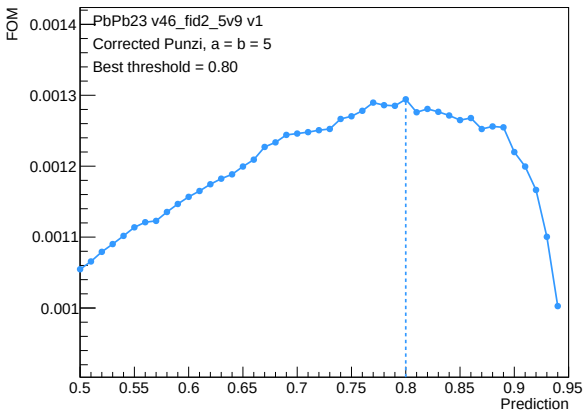
Input variables

Btrk1dR, Btrk2Pt
BtrkPtimb, Bmu1pt
Bmu2pt

X MC: All pthat components retained.
pThatreweight; no extra reweighting.
No additional signal training mass window.

DATA training background
$(3.80, 3.85) \cup (3.89, 3.94)$ GeV.
Unit weights; same common selection.

Five variables, all pthat: Punzi FOM

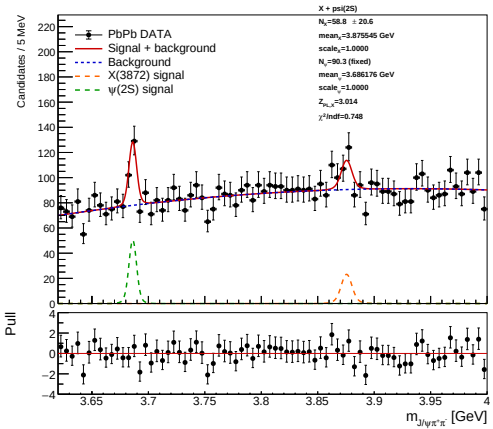


Empirical corrected Punzi
$a = b = 5$.

Score: 0.50–0.95
Step: 0.01

Grid maximum
score > 0.80
FOM $= 1.294304 \times 10^{-3}$

Five variables, all pthat: optimal two-peak fit



score > 0.80

$N_X = 58.83 \pm 20.60$
$N_{\psi(2S)} = 90.33$ (fixed)

$\sqrt{q_0} = 3.014$

MC double-Gaussian widths fixed.
2nd-order Chebyshev background.

Six variables, pthat5 excluded: training

Common training selection

Variable	Selection before ML
Bpt	$15 < \texttt{Bpt} < 50$ GeV
By	$\lvert\texttt{By}\rvert < 1.6$
BQvalue	$\texttt{BQvalue} < 0.15$ GeV
Btrk2dR	$\texttt{Btrk2dR} < 0.35$

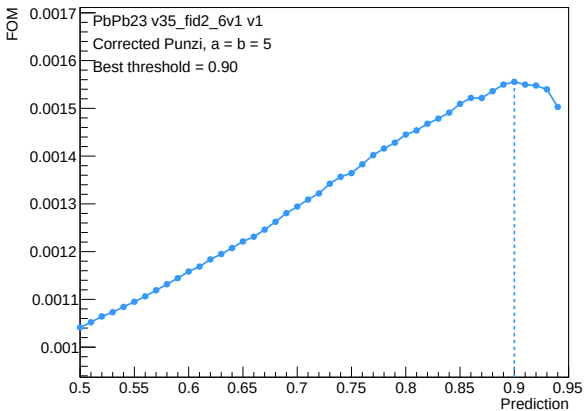
Input variables

Btrk1dR, Btrk2Pt
BtrkPtimb, Bcos_dtheta
Bmu1pt, Bmu2pt

X MC: pthat5 excluded in training and application.
pThatreweight; no extra reweighting.
No additional signal training mass window.

DATA training background
$(3.80, 3.85) \cup (3.89, 3.94)$ GeV.
Unit weights; same common selection.

Six variables, pthat5 excluded: Punzi FOM

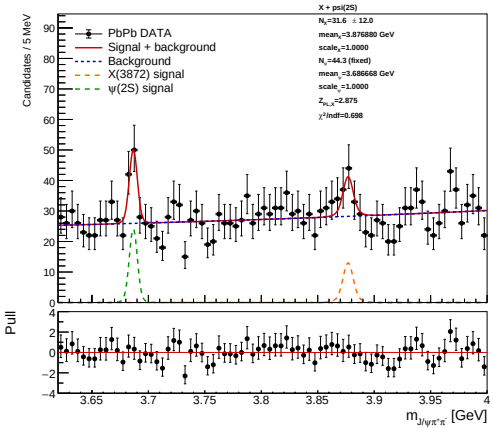


Empirical corrected Punzi
$a = b = 5$.

Score: 0.50–0.95
Step: 0.01

Grid maximum
score > 0.90
FOM $= 1.555545 \times 10^{-3}$

Six variables, pthat5 excluded: optimal two-peak fit



score > 0.90

$N_X = 31.62 \pm 11.97$
$N_{\psi(2S)} = 44.33$ (fixed)

$\sqrt{q_0} = 2.875$

MC double-Gaussian widths fixed.
2nd-order Chebyshev background.

Five variables, pthat5 excluded: training

Common training selection

Variable	Selection before ML
`Bpt`	$15 < \texttt{Bpt} < 50$ GeV
`By`	$\lvert\texttt{By}\rvert < 1.6$
`BQvalue`	$\texttt{BQvalue} < 0.15$ GeV
`Btrk2dR`	$\texttt{Btrk2dR} < 0.35$

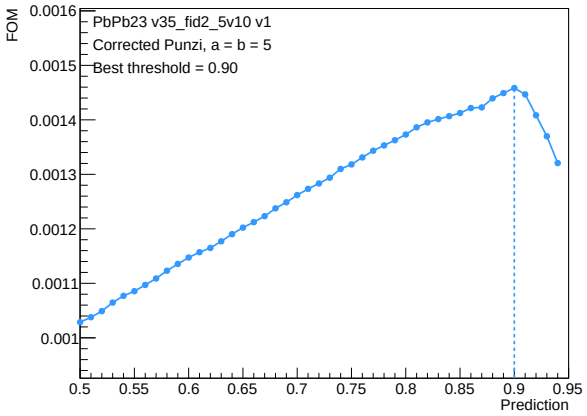
Input variables

`Bmu1pt`, `Bmu2pt`
`Btrk2Pt`, `BtrkPtimb`
`Btrk1dR`

X MC: pthat5 excluded in training and application.
`pThatreweight`; no extra reweighting.
No additional signal training mass window.

DATA training background
$(3.80, 3.85) \cup (3.89, 3.94)$ GeV.
Unit weights; same common selection.

Five variables, pthat5 excluded: Punzi FOM

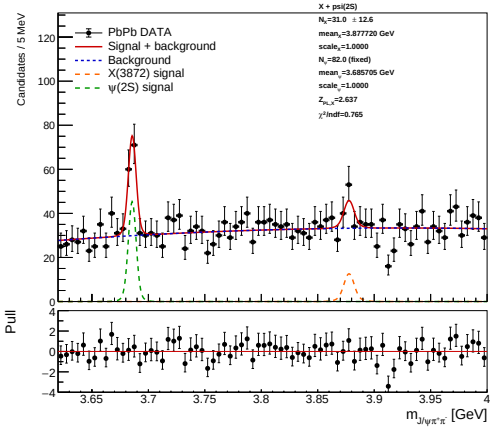


Empirical corrected Punzi
$a = b = 5$.

Score: 0.50–0.95
Step: 0.01

Grid maximum
score > 0.90
FOM $= 1.458448 \times 10^{-3}$

Five variables, pthat5 excluded: optimal two-peak fit



score > 0.90

$N_X = 30.96 \pm 12.62$
$N_{\psi(2S)} = 82.02$ (fixed)

$\sqrt{q_0} = 2.637$

MC double-Gaussian widths fixed.
2nd-order Chebyshev background.